Exp:02 Creat an Application using ipywidgets

Aim:-

To create an simple python application using ipywidg library in python.

Libraries Used:-

* ipy Widgets.
* IPython.display
* base64
* os
* mimetypes
* datetime.

Procedure:-

* Import required libraries and load college logo as base string for Water Marking.
* Define reference data-blocks, floors and dept wise complain Category map to their routing mail ids.
* Build tab1 widgets for capture complainant location de
* Build tab2 widgets for capture issue with proof
* Implement Dynamic behaviour - category change upd issue list and routed email.
* Write the Submit handler to valid inputs, bind a HT Summary with watermark and proof.
* Assemble all tabs under a titled "Tab" widget and display the complete application.

Code:-

```
def get_logo_base64():
    if os.path.exists(LOGO_PATH):
        with open(LOGO_PATH, "rb") as f:
            return base64.b64encode(f.read()).decode("utf-8")
    return None
LOGO_B64 = get_logo_base64()

CATEGORIES = {
    "Facility - (FAN / Light / Smartboard / Chalk / Restroom)": {
        "email": "sec-sections@gmail.com",
        "items": ["Fan not working" etc...]
}  }
def _on_category_change(change):
    cat = change["new"]
    item_field.options = CATEGORIES[cat]["items"]
    dept_email_disp.value = f"Routed to: {CATEGORIES[cat]
["email"]}"
category_field.observe(on_category_change, name="value")
def on_item_change(change):
    other_row.layout.display = "flex" if change["new"]=="Other"
    else "none"
item_field.observe(on_item_change, name="value")
def make_remove_handler(file_index):
    def handler():
        proof_files.pop(file_index)
        render_proof_preview()
    return handler
```

Output:

Complaint Box - REC

1. Your Details | 2. Issue & Submit | Complaint Log

Step 2: Issue Details

Category: Facility (Fan/Light/Smartboard/Chalk/Restroom)

Issue/Item: Fan not working

Details: Fan switch is not working

Upload photo/video proof (optional)

Upload Proof (0)

image1 (1).jpg Remove



This complaint will be routed to: rec-ecbikes@gmail.com

Submit Complaint

Complaint Submitted

Name: Mohamed Ashiaq

Email: mohamedashiaq2018@gmail.com

Block: A | Floor: Floor 1 | Room: 101

Category: Facility (Fan/Light/Smartboard/Chalk/Restroom)

Issue/Item: Fan not working

Details: Fan switch is not working

Routed to: rec-ecbikes@gmail.com

Submitted at: 2026-07-24 11:27:05

Proof:

image1 (1).jpg



31/7/26

Result:-

Created an simple application on python successfully using ipywidgets.